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Cmpsc 132

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4/28/26

Cmpsc 132 Final Report

The game I decided to make was the number guessing game. The goal of the program is to have the computer generate a random number within 1-100 (inclusive), and a player attempts to guess that number. They are given either a “Too high” or “Too low” feedback with each incorrect guess. There are many programming concepts we learned throughout the course implemented into the game.

The game first is built by generating the random number by using Python’s random module. A while loop is used to continuously ask the player for an input until it is guessed correctly. The user input is allowed by using the input() function. I also implemented a piece to handle ValueError. This is there so the program won’t end if the player’s unput was invalid.

With a valid user guess I then used conditional statements to compare the input with the randomly generated number and would provide feedback. The player will use that to help for their next guess. Once the correct number is guessed, the program will display the total number of attempts.

The project I built contains important programming concepts including functions, loops, conditionals, exception error handling, and user input. Then there is a logical aspect that gives a guide to the player toward the correct number. One challenge for me was trying

to find an enhancement to implement. Throughout my testing of the basic program I thought of a possible invalid input the user could give. So, then I took time to correctly place in the try, except handle and it worked well.

```
I'm thinking of a random number between 1 and 100
Enter a guess: fifty
Enter a valid number
Enter a guess: 50
Too low
Enter a guess: 75
Too low
Enter a guess: 87
Too low
Enter a guess: 95
Too high
Enter a guess: 92
You are correct in 5 attempts.
```